

The scaled logarithmic derivative of θ_4 is decreasing and convex

Candidate full solution to an assertion in arXiv:1601.02972

11 August 2026

Status: candidate full result; likely valid; human review needed. For every $s > 0$, the function

$$H(s) = s^2 \frac{\theta'_4(s)}{\theta_4(s)}$$

is strictly decreasing and strictly convex.

Source assertion

Faulhuber and Steinerberger define

$$\theta_4(s) = \sum_{k \in \mathbb{Z}} (-1)^k e^{-\pi k^2 s}, \quad s > 0.$$

After their refined logarithmic concavity theorem, they state that the stronger claim that $s^2\theta'_4(s)/\theta_4(s)$ is decreasing and convex appears true, but that their methods do not show it [1, p. 3]. The exact source text is reproduced below.

$\{s \in \mathbb{R} : s \geq 1\}$ and then use the new identity (which follows from the Jacobi identity)

$$s \frac{\theta'_3(s)}{\theta_3(s)} + \frac{1}{s} \frac{\theta'_3(\frac{1}{s})}{\theta_3(\frac{1}{s})} = -\frac{1}{2} \quad \text{for } s > 0$$

to backpropagate the information to the origin.

Theorem 2.4 (Refined logarithmic concavity for θ_4). *We have, for $s > 0$,*

$$\theta''_4(s)\theta_4(s) - \theta'_4(s)^2 < -\frac{\theta'_4(s)\theta_4(s)}{s} < 0.$$

This inequality is new (however, see Coffey & Csordas [4] for a related function). The proof is based on exploiting a suitable monotonicity property using an infinite product representation of θ_4 . Since θ_4 has alternating signs, it is difficult to handle and the proof contains a certain 'magic' element of algebraic simplification. Interestingly, this argument does not work for θ_3 even though there exists an analogous representation of θ_3 as an infinite product. Theorem 2.3 implies that $s\theta'_3(s)/\theta_3(s)$ is monotonically increasing on $\mathbb{R}^+$ while Theorem 2.4 yields that $s\theta'_4(s)/\theta_4(s)$ is monotonically decreasing. Stronger results seem to be true: $s^2\theta'_4(s)/\theta_4(s)$ seems to be monotonically decreasing and convex but our arguments cannot show that.

Theorem and proof intuition

Theorem 1. *The function $H(s) = s^2\theta'_4(s)/\theta_4(s)$ satisfies*

$$H'(s) < 0 \quad \text{and} \quad H''(s) > 0 \quad (s > 0).$$

Proof intuition. The two difficult ends of the parameter range want different expansions. For $s \geq 1$, Jacobi's product expresses H as a positive sum of copies of $x^2/(e^x - 1)$ evaluated at $x \geq \pi$; every copy is decreasing and convex. For $s \leq 1$, modular inversion exposes the leading behavior $\pi/4 - s/2$. Its remaining terms can be paired as $e^{-C/s} - 3e^{-D/s}$ with $D > C$ and $D - C \geq 2\pi$. The exponential separation makes every pair increasing and convex. Their total slope is nevertheless less than 0.32, so it cannot overcome the leading slope $-1/2$; their positive curvatures establish convexity.

Two elementary estimates

Set

$$\phi(x) = \frac{x^2}{e^x - 1}.$$

Lemma 1. $\phi'(x) < 0$ and $\phi''(x) > 0$ for every $x \geq \pi$.

Proof. Direct differentiation gives

$$\phi'(x) = \frac{x((2-x)e^x - 2)}{(e^x - 1)^2} < 0 \quad (x \geq \pi > 2).$$

The numerator of $(e^x - 1)^3 \phi''(x)$ is

$$N(x) = (x^2 - 4x + 2)e^{2x} + (x^2 + 4x - 4)e^x + 2.$$

Moreover

$$N'(x) = xe^x((2x - 6)e^x + x + 6) > 0 \quad (x \geq \pi > 3).$$

It remains only to check the left endpoint without a decimal computation. The elementary bounds $157/50 < \pi < 22/7$ and $e^\pi < 24$ imply, since $\pi^2 - 4\pi + 2 < 0$,

$$\begin{aligned} e^{-\pi} N(\pi) &> (24)(\pi^2 - 4\pi + 2) + (\pi^2 + 4\pi - 4) \\ &= 25\pi^2 - 92\pi + 44 > 25(157/50)^2 - 92(157/50) + 44 > 0. \end{aligned}$$

(For completeness, $e^\pi < 24$ follows from $\pi < 3 + 1/7$, $e^3 < 20.1$, and $e^{1/7} < 7/6$.) Thus $N(x) > 0$ on the required range. $\square$

The large-parameter half

Jacobi's product is

$$\theta_4(s) = \prod_{n \geq 1} (1 - e^{-2\pi ns})(1 - e^{-(2n-1)\pi s})^2.$$

Logarithmic differentiation yields

$$H(s) = s^2 \left[2\pi \sum_{n \geq 1} \frac{n}{e^{2\pi ns} - 1} + 2\pi \sum_{n \geq 1} \frac{2n-1}{e^{(2n-1)\pi s} - 1} \right]. \quad (1)$$

Every summand in (1) is a positive constant times $\phi(cs)$, with $c \geq \pi$. Hence Lemma 1 permits termwise differentiation and gives

$$H'(s) < 0, \quad H''(s) > 0 \quad (s \geq 1). \quad (2)$$

The small-parameter half

The imaginary transformation and the product for θ_2 give, for $t > 0$,

$$\theta_4(s) = s^{-1/2}\theta_2(1/s), \quad \theta_2(t) = 2e^{-\pi t/4} \prod_{n \geq 1} (1 - e^{-2\pi nt})(1 + e^{-2\pi nt})^2.$$

After logarithmic differentiation and expansion as geometric series,

$$H(s) = \frac{\pi}{4} - \frac{s}{2} + R(s), \quad R(s) = 2\pi \sum_{n \geq 1} n \sum_{j \geq 0} \left(e^{-C_{n,j}/s} - 3e^{-D_{n,j}/s} \right), \quad (3)$$

where

$$C_{n,j} = 2\pi n(2j+1), \quad D_{n,j} = 2\pi n(2j+2).$$

Indeed, if $a = 2\pi n/s$, then the relevant bracket is

$$\frac{2}{e^a + 1} - \frac{1}{e^a - 1} = \frac{e^a - 3}{e^{2a} - 1} = \sum_{j \geq 0} \left(e^{-(2j+1)a} - 3e^{-(2j+2)a} \right).$$

Fix $0 < s \leq 1$ and abbreviate $C = C_{n,j}$, $D = D_{n,j}$. For $P(s) = e^{-C/s} - 3e^{-D/s}$,

$$P'(s) = \frac{Ce^{-C/s}}{s^2} \left[1 - 3\frac{D}{C}e^{-(D-C)/s} \right], \quad (4)$$

$$P''(s) = \frac{C(C-2s)e^{-C/s}}{s^4} \left[1 - 3\frac{D(D-2s)}{C(C-2s)}e^{-(D-C)/s} \right]. \quad (5)$$

Let $q = e^{-2\pi} < 1/500$. Here $D/C \leq 2$ and $D - C = 2\pi n$, so the bracket in (4) is at least $1 - 6q > 0$. Also

$$\frac{D(D-2s)}{C(C-2s)} \leq 2 \left(2 + \frac{2}{2\pi - 2} \right) < 5,$$

so the bracket in (5) is at least $1 - 15q > 0$. Consequently every pair is strictly increasing and strictly convex, and therefore

$$R'(s) > 0, \quad R''(s) > 0. \quad (6)$$

It remains to keep the positive slope of R below $1/2$. Dropping the negative member of every differentiated pair and observing that $s^{-2}e^{-C/s}$ increases on $(0, 1]$, we obtain

$$\begin{aligned} R'(s) &< 4\pi^2 \sum_{n \geq 1} n^2 \sum_{j \geq 0} (2j+1)q^{n(2j+1)} \\ &\leq 4\pi^2 \sum_{n \geq 1} \frac{n^2 q^n}{(1-q^n)^2} \leq 4\pi^2 \frac{q(1+q)}{(1-q)^5} < 0.32 < \frac{1}{2}. \end{aligned}$$

In the last step we used $\pi^2 < 10$, $q < 1/500$, $1+q < 2$, and $(1-q)^{-5} < 2$. Equations (3) and (6) now give

$$H'(s) = -\frac{1}{2} + R'(s) < 0, \quad H''(s) = R''(s) > 0 \quad (0 < s \leq 1).$$

Together with (2), this proves Theorem 1. □

Scope and novelty note

This theorem proves only the precise theta-function assertion on source PDF page 3. It does not resolve the paper’s much broader Strohmer–Beaver hexagonal-lattice conjecture. The run’s four cheap indexes and targeted arXiv searches for the exact scaled logarithmic derivative statement found the source paper but no later primary-source resolution. Novelty therefore remains provisional pending expert literature review.

References

- [1] M. Faulhuber and S. Steinerberger, *Optimal Gabor frame bounds for separable lattices and estimates for Jacobi theta functions*, arXiv:1601.02972v3 (2016).